📘 **AI/ML Internship – Day 9 Notes & Tasks**

🟢 **Module 3: Python for AI**

📌 **Part 1: Real-World Problem Solving Approach**

🔹 **How Real Projects Work?**

In real AI applications:

- Data comes from files
- Data is processed using logic
- Output is generated

👉 So today we combine:

- File Handling
- Loops
- Functions
- Data Structures

📌 **Part 2: Working with Data from Files**

🔹 **Example File Content (students.txt)**

```
John,80
Sara,90
Alex,70
```

🔹 **Reading and Processing Data**

```
students = []

with open("students.txt", "r") as file:
    for line in file:
        name, marks = line.strip().split(",")
        students.append({"name": name, "marks": int(marks)})

print(students)
```

🔹 Finding Average Marks

```
total = sum(s["marks"] for s in students)
avg = total / len(students)
print("Average:", avg)
```

🔹 Finding Topper

```
top = max(students, key=lambda x: x["marks"])
print("Topper:", top["name"])
```

📌 Part 3: Writing Processed Data to File

```
with open("output.txt", "w") as file:
    for s in students:
        file.write(s["name"] + "\n")
```

📌 Part 4: Error Handling in Data Processing

```
try:
    with open("students.txt", "r") as file:
        data = file.read()
except FileNotFoundError:
    print("File not found")
```

📌 Part 5: Functions for Data Processing

```
def get_average(students):
    return sum(s["marks"] for s in students) / len(students)

print(get_average(students))
```

🎯 Tasks for Day 9

🧠 Task 1: Theory

1. How data is handled in real-world applications?
2. Why file handling is important in AI?
3. What is data processing?
4. Why use functions in data processing?
5. What is error handling in files?

💻 **Task 2: File Data Practice**

1. Create file with student data
2. Read and store in list of dictionary
3. Print all students
4. Find average marks
5. Find topper

💻 **Task 3: Data Processing Programs**

1. Filter students with marks > 75
2. Count number of students
3. Find lowest marks
4. Sort students by marks

💻 **Task 4: Function Practice**

1. Function to calculate average
2. Function to find topper
3. Function to filter students

🔥 **Task 5: Logic Challenge**

1. Find student with second highest marks
2. Count number of passed/failed students
3. Convert data into dictionary format